Efficient Quantum PDE Solvers and Gibbs Samplers on the Continuum via Gevrey Regularity

Pooya Ronagh

Pr. Quantum System Architecture | Microsoft Quantum

References for this talk

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- **Gibbs Sampling of Periodic Potentials on a Quantum Computer**, arXiv:2210.08104
  with Arsalan Motamedi
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- **Breaking the Curse of Dimensionality in Quantum PDE Solvers via Gevrey Regularity**, arXiv:2608.07893
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- **Provable Quantum—Classical Separation for Continuous Gibbs Sampling**, arXiv:2608.24527
  with Enrico Olivucci, Mariia Sobchuk, Sehmimul Hoque, Jeffrey Hnybida, Kyungho W. Kim, Ala Shayeghi

Motivation: solving general linear PDEs on quantum computer

$$\sum_{\alpha \in \mathcal{J}} g_\alpha(x) D^\alpha u(x) = \eta(x)$$

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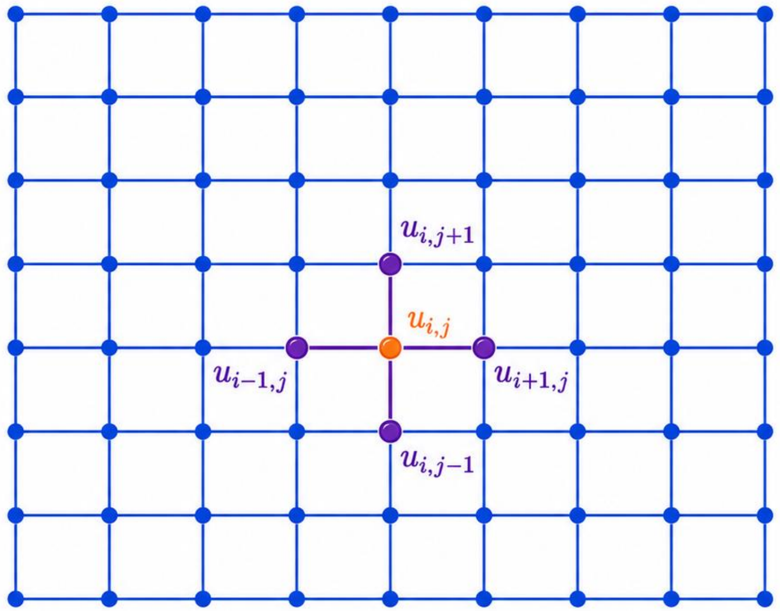
- The goal is to linearize the generator and obtain a large linear system:

$$\mathcal{L} = \sum_{\alpha \in \mathcal{J}} g_\alpha(x) D^\alpha \longrightarrow \mathbb{L}_N \left|u_N\right\rangle = \left|\eta_N\right\rangle$$



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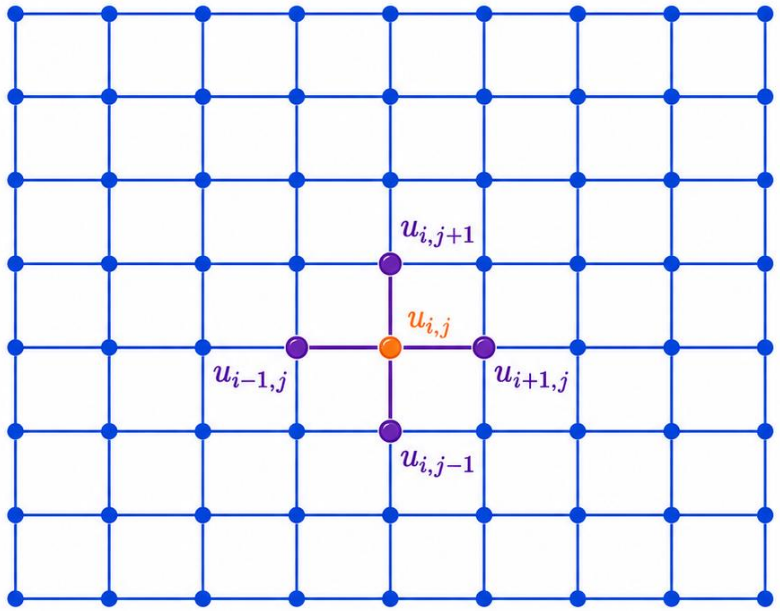

- Problem: Grid cells have to be as small as $1/\epsilon$.
- Solution: Hope for a "smooth" behaviour and use planewave basis (i.e., move to the Fourier domain).

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- What we can actually do:

$$\widetilde{D}^\alpha = (F_N^{\otimes d})^{-1}\,\mathrm{diag}\left(\left\{\prod_{j=1}^{d}(i\omega_j)^{\alpha_j}\right\}_{\omega\in\Lambda_N}\right)F_N^{\otimes d}$$

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A Fourier analytic crash course

$$f(x) = \int_{\mathbb{R}^d} \hat{f}(\omega)\, e^{i\langle \omega, x\rangle}\, d\omega$$

$$\hat{f}(\omega) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} f(x)\, e^{-i\langle \omega, x\rangle}\, dx$$

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$f(x) = \sum_{\omega\in\mathbb{Z}^d} \widehat{f}_\omega\, e^{i\langle \omega, x\rangle}$	Fourier transform / series
$\widehat{f}_\omega = \frac{1}{(2\pi)^d}\int_{\mathbb{T}^d} f(x) e^{-i\langle \omega, x\rangle}\, dx = \mathbb{E}_{\mathbb{T}^d}[f(x)e^{-i\langle \omega, x\rangle}]$	Fourier coefficients
$\widetilde{f}_\omega = \mathbb{E}_{\Gamma_N}[f(x)e^{-i\langle \omega, x\rangle}]$	Discrete Fourier coefficients
$\lvert\widetilde{f}_N\rangle = \frac{1}{\sqrt{(2N+1)^d}}\sum_{\omega\in\Lambda_N} \widetilde{f}_\omega\, \lvert\omega\rangle = F_N^{\otimes d}\lvert f_N\rangle$	Discrete / quantum Fourier transform
$\widetilde{f}_\omega = \sum_{m\in\mathbb{Z}^d} \widehat{f}_{\omega+(2N+1)m}$	Aliasing sums

For periodic functions

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$$|f_N\rangle := \frac{1}{\sqrt{\langle\!\langle f^2 \rangle\!\rangle_{\Gamma_N}}} \sum_{x \in \Gamma_N} f(x)\,|x\rangle \xrightarrow{F_N^{\otimes d}} |\widetilde{f}_N\rangle := \frac{1}{\sqrt{\langle f^2 \rangle_{\Gamma_N}}} \sum_{\omega \in \Lambda_N} \widetilde{f}_\omega\,|\omega\rangle$$

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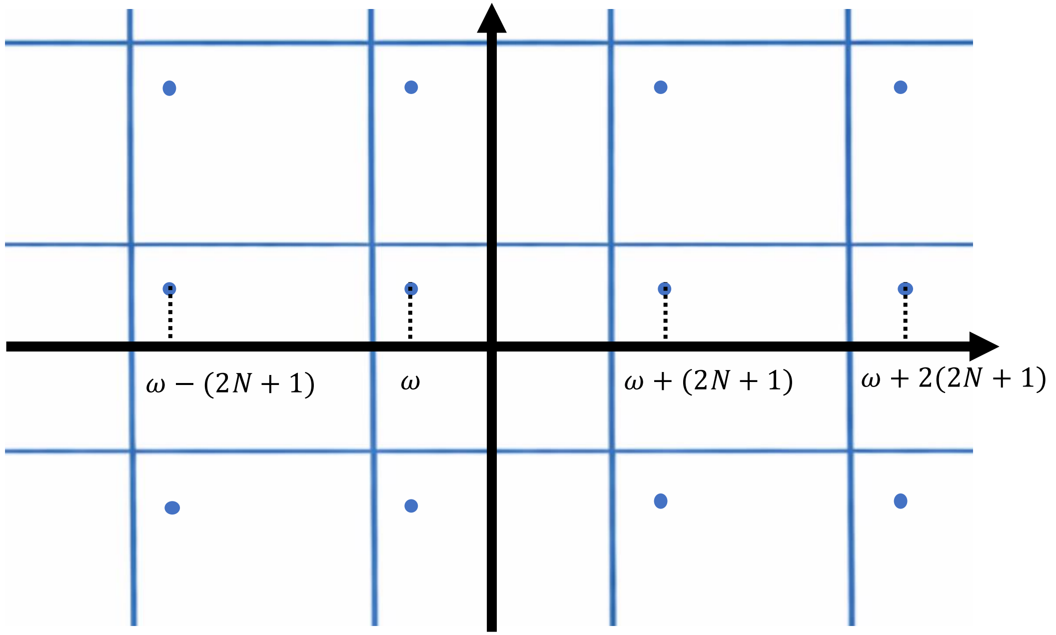

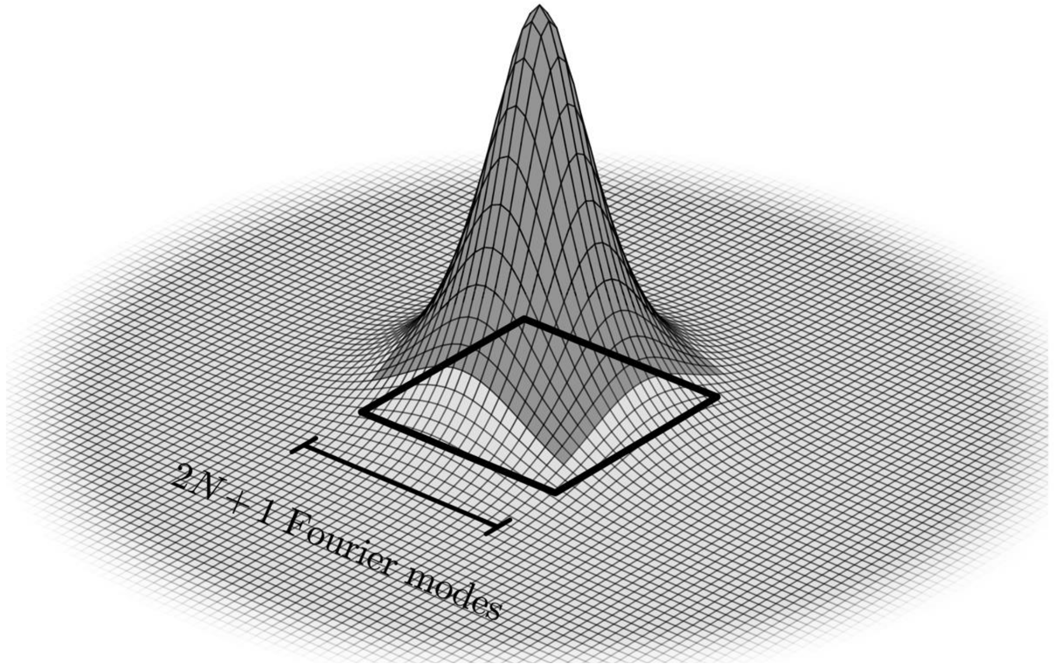

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Regularity class	Decay rate of Fourier coefficients ($\lvert\widehat{f_\omega}\rvert$)	Tailedness at cutoff N (truncation error, E_N)	Cutoff N to guarantee target precision ε
Analytic ($\mathcal{C}^\omega$ or $\mathcal{G}^1$)	$Ce^{-r\lVert\omega\rVert_\infty}$	$C\sqrt{\frac{d}{r}N^{d-1}}\, e^{-rN}$	$\frac{1}{2r}\log\frac{C^2 d}{r\varepsilon}$

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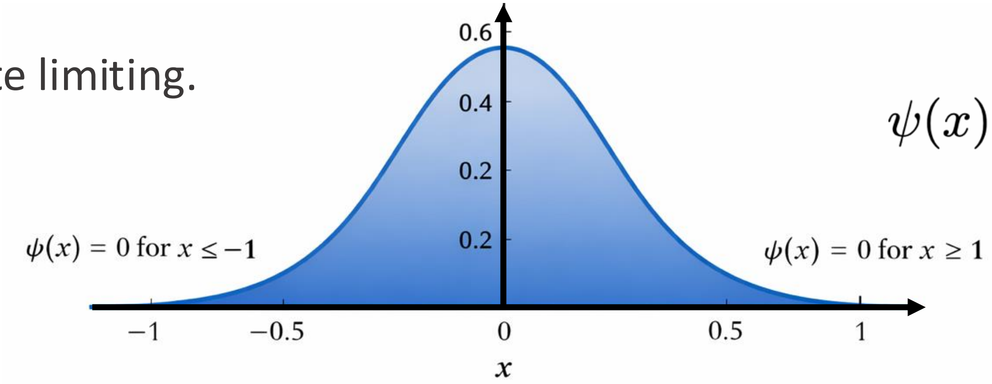
- But analyticity is quite limiting.



$$\psi(x) = \begin{cases} \exp\left(-\frac{1}{1-x^2}\right), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Regularity class	Decay rate of Fourier coefficients ($\lvert\widehat{f}_\omega\rvert$)	Tailedness at cutoff N (truncation error, E_N)	Cutoff N to guarantee target precision ε
Continuous ($\mathcal{C}^0$)	$o(1)$	$o(1)$	none
p-smooth ($\mathcal{C}^p$)	$C\lVert\omega\rVert_\infty^{-p}$	$C\sqrt{\frac{d}{d-2p}N^{-p+d/2}}$	$\left(\frac{C}{\varepsilon}\right)^{1/(p-d/2)}$
(p,q)-Hölder ($\mathcal{C}^{p,q}$)	$C\lVert\omega\rVert_\infty^{-p-q}$	$C\sqrt{\frac{d}{d-2p-2q}N^{-p-q+d/2}}$	$\left(\frac{C}{\varepsilon}\right)^{1/(p+q-d/2)}$
smooth ($\mathcal{C}^\infty$)	sub-polynomial	sub-polynomial	super-logarithmic
s-Gevrey ($\mathcal{G}^s$ for $s>1$)	$Ce^{-r\lVert\omega\rVert_\infty^{1/s}}$	$C\sqrt{\frac{ds}{r}N^{d-1/s}}\,e^{-rN^{1/s}}$	$\left(\frac{1}{2r}\,\log\frac{C^2ds}{r\varepsilon}\right)^s$
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Entire ($\mathcal{E}$)	faster than $Ce^{-r\lVert\omega\rVert_\infty}$, $\forall\, r>0$	faster than Ce^{-rN}, $\forall\, r>0$	sub-linear in $\log\frac{1}{\varepsilon}$
Entire of finite order ($\mathcal{G}^s$ for $0\le s<1$)	$Ce^{-r\lVert\omega\rVert_\infty^{1/s}}$	$C\sqrt{\frac{ds}{r}N^{d-1/s}}\,e^{-rN^{1/s}}$	$\left(\frac{1}{2r}\,\log\frac{C^2ds}{r\varepsilon}\right)^s$
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Gevrey regularity and the answer to our burning question

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SUR LA

NATURE ANALYTIQUE DES SOLUTIONS

DES

ÉQUATIONS AUX DÉRIVÉES PARTIELLES

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- which means that for s-Gevrey functions

$$\left\| \widetilde{D}_N^{\alpha} f_N - (D^{\alpha} f)_N \right\|_2 \leq C e^{-\frac{r}{2} N^{1/s}}$$

$$\left\| \widetilde{f}_N - \widehat{f} \right\|_2 \leq C e^{-\frac{r}{2} N^{1/s}}$$

- provided choosing $N \geq \left(\frac{(d+|\alpha|)s}{r}\right)^s$.

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- Total gate count: $\widetilde{\mathcal{O}}\left(\frac{\lceil g \rceil \, |\mathcal{J}| \, (|\mathcal{J}| + \lceil \mathcal{J} \rceil)}{\sigma_{\min}^{+}} d^{\lceil \mathcal{J} \rceil s+1} \left(s \log\left(\frac{1}{\|\widehat{u}\| \, \varepsilon}\right)\right)^{\lceil \mathcal{J} \rceil s} \log\left(\frac{1}{\varepsilon}\right)\right)$

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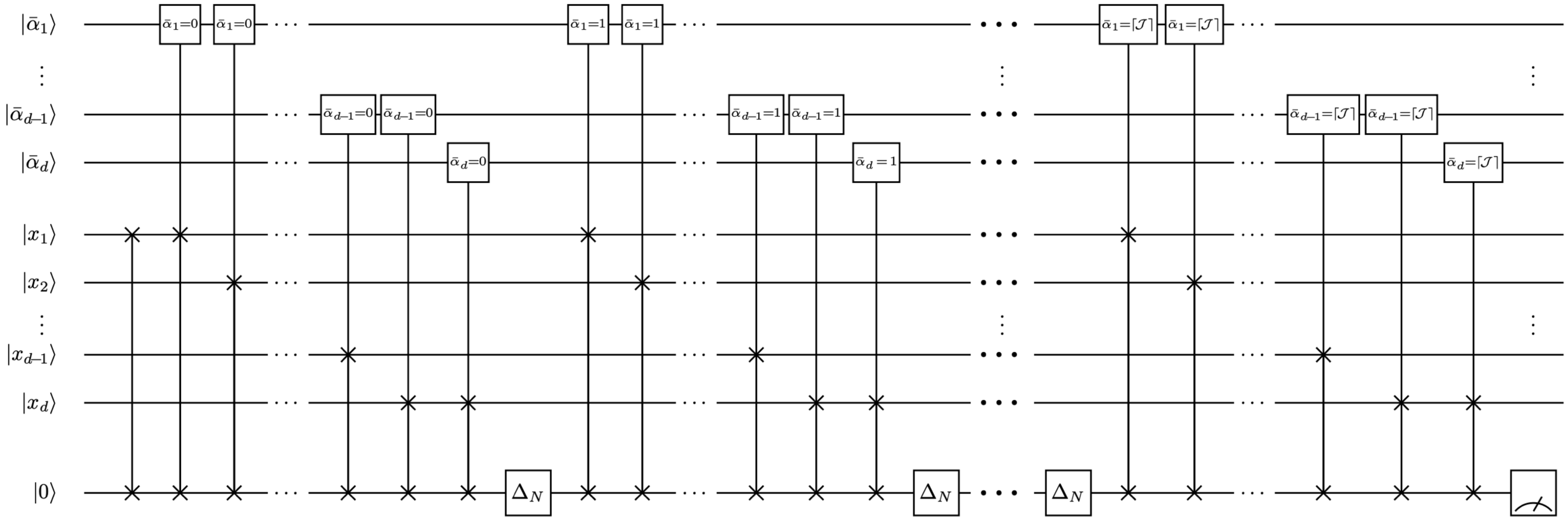
- Total qubit count: $\mathcal{O}(d(\log N + \log \lceil \mathcal{J} \rceil))$

Block encoding of the approximate differentials



$$\Delta_N = F_N^{-1}\,\mathrm{diag}\{(i\omega)_{\omega\in\{-N,\cdots,N\}}\}F_N$$

$O(N^2 \log N)$ elementary gates
$O(\log N)$ qubits

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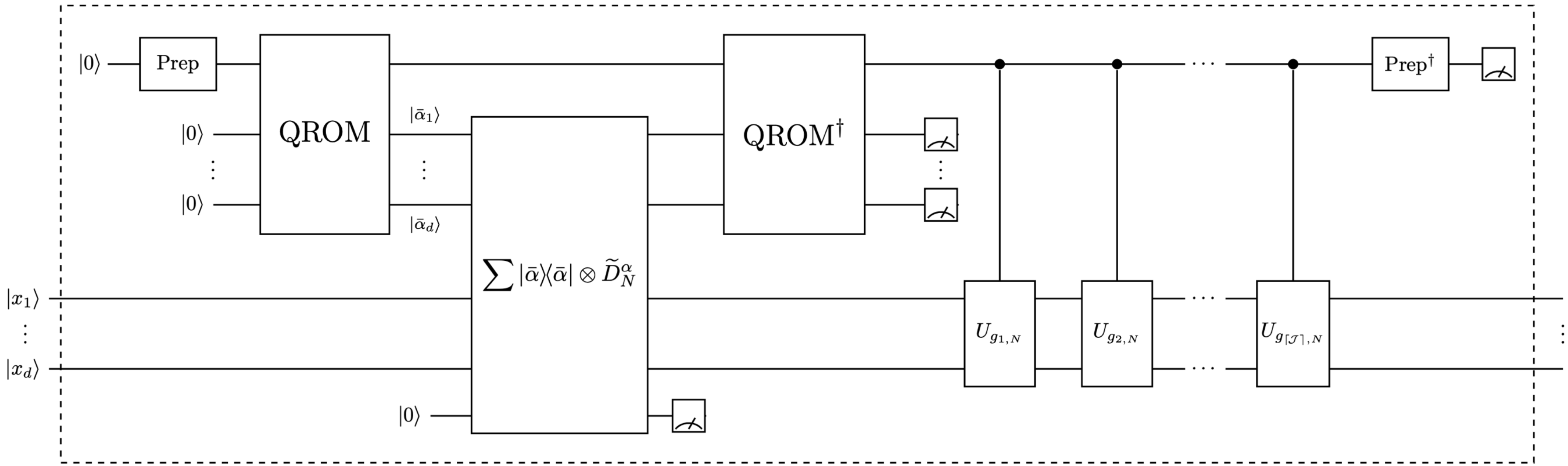
Block encoding of $\mathbb{L}_N$



$\mathbb{L}_N$

Breaking the curse of dimensionality

Algorithm	Mathematical assumptions	Complexity in ε and d	Reference
Uniform-grid FDM/FEM/FVM	Finitely differentiable	$\mathrm{poly}((1/\varepsilon)^d)$	folklore
Adaptive h–p-FEM on geometrically graded meshes	Analytic on a polygonal domain with possibly finitely many corner singularities	$\mathrm{poly}(\log^d(1/\varepsilon))$	[BS87, Sec 6]
Spectral methods on full tensor grids	Isotropically analytic in each variable (Bernstein ellipse / periodic strip)	$\mathrm{poly}(\log^d(1/\varepsilon))$	[Tad86, Thms 3.1, 4.4] [Ghe07, Rmk 27] [STW11, Rmk 1.10]
Sparse-grid FEM	Bounded mixed second derivatives (dominating mixed smoothness)	$\varepsilon^{-1/2}\log^{3(d-1)/2}(1/\varepsilon)$	[Zen91, Eq 2.3, 2.9] [BG04, Thm 3.8, Lem 3.13]
Sparse spectral methods (hyperbolic cross)	Bounded mixed derivatives up to a fixed order m (Jacobi-weighted Korobov space)	$\varepsilon^{-1/m}\log^{d-1}(1/\varepsilon)$	[STW11, Thm 8.10]
This work	Gevrey regularity ($s \geq 0$)	$(sd)^{\lceil \mathcal{J} \rceil s}\log^{\lceil \mathcal{J} \rceil s+1}(\frac{1}{\varepsilon})$	Theorem 5

Application 1: Atomistic simulation $H|\psi\rangle = |\eta\rangle$

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$$H = -\frac{1}{2}\sum_{i=1}^{M} \Delta_{x_i} + V(x), \quad \text{where} \quad V(x) = -\sum_{i=1}^{M}\sum_{k=1}^{P} \frac{Z_k}{r_{ik}} + \sum_{1\le i<j\le M} \frac{1}{r_{ij}}$$

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- Challenge: The solution wavefunction is cuspidal (Kato cusps).
- Common practice:

$$V_\gamma(x) = -\sum_{i=1}^{M}\sum_{k=1}^{P}\frac{Z_k}{\sqrt{r_{ik}^2+\gamma^2}} + \sum_{1\le i<j\le M}\frac{1}{\sqrt{r_{ij}^2+\gamma^2}}$$

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- Total gate count: $$\widetilde{\mathcal{O}}\left(\frac{M^7(PZ+M)(P+M)^3}{\sigma_{\min}^{+}(\mathbb{L}_N)\,(E_1-E_0)^6\,\varepsilon^6}\log^2\left(\frac{1}{\varepsilon\,\|\delta\widehat{\Psi}\|}\right)\right)$$
- Total qubit count: $$\mathcal{O}\left(\frac{M^3(P+M)}{(E_1-E_0)^2\varepsilon^2}\log\left(\frac{1}{\varepsilon\left\|\delta\widehat{\Psi}\right\|}\right)\right)$$

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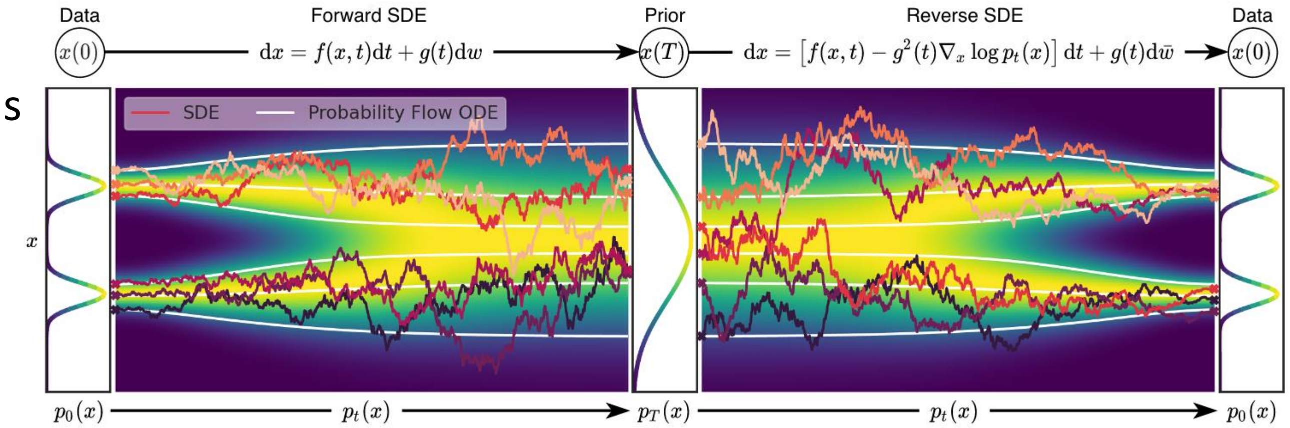
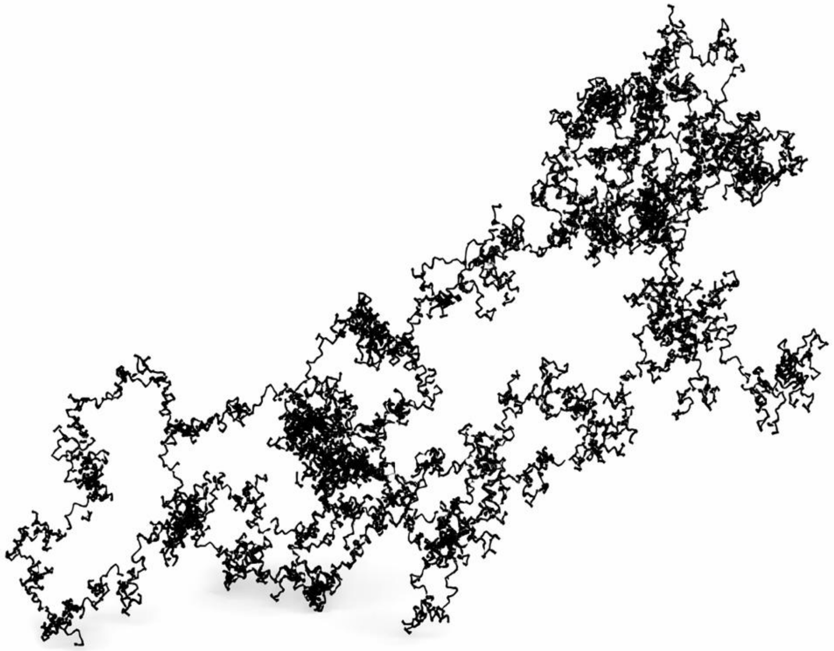
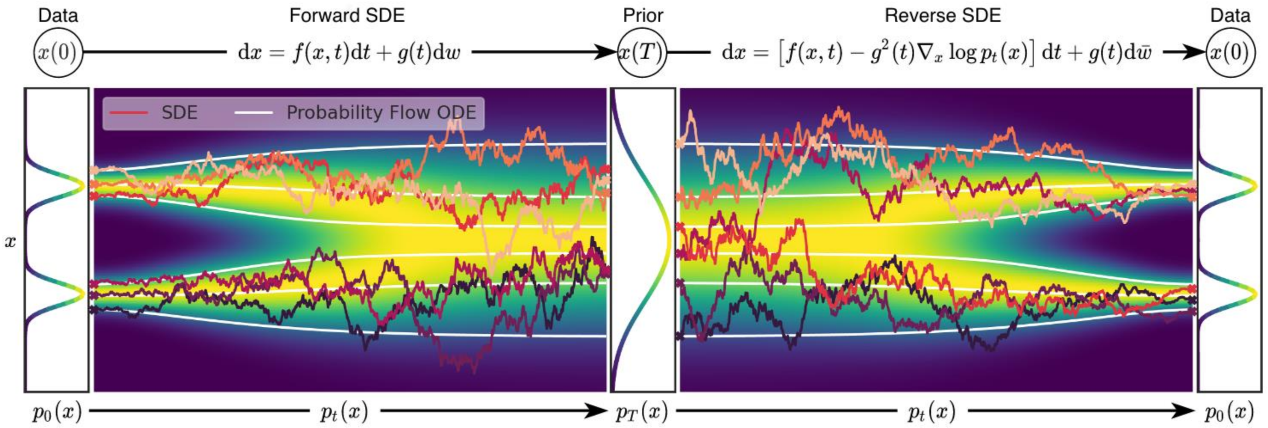


Application 2: Provable separation for quantum Gibbs sampling

- Focus: Gibbs sampling on the continuum
- Common in ML: Diffusion models and EBMs
- Method: Simulating **Langevin dynamics**

$$dX_t = -\frac{1}{2}\beta\sigma^2\nabla_X E_\theta(X)\,dt + \sigma\,dW_t.$$





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- Feynman–Kac formula / forward and backward Kolmogorov equations

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- Transition operators $P(t,x,dy) = p(t,x,y)dy$ provides detailed balance.

$$\frac{\partial}{\partial t}p(.,.,y) = \mathcal{L}_x p(.,.,y) \qquad \frac{\partial}{\partial t}p(.,x,.) = \mathcal{L}_y^* p(.,x,.)$$

Pooya Ronagh | University of Waterloo | Microsoft Quantum

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- Parabolic Fokker–Planck equation $\frac{\partial}{\partial t}p = \mathcal{L}p$ has $p \propto e^{-\beta E}$ as steady state solution.

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- OSHHKSR26: Fixed all the above caveats for Gevrey potentials.

Eliminating the warm start

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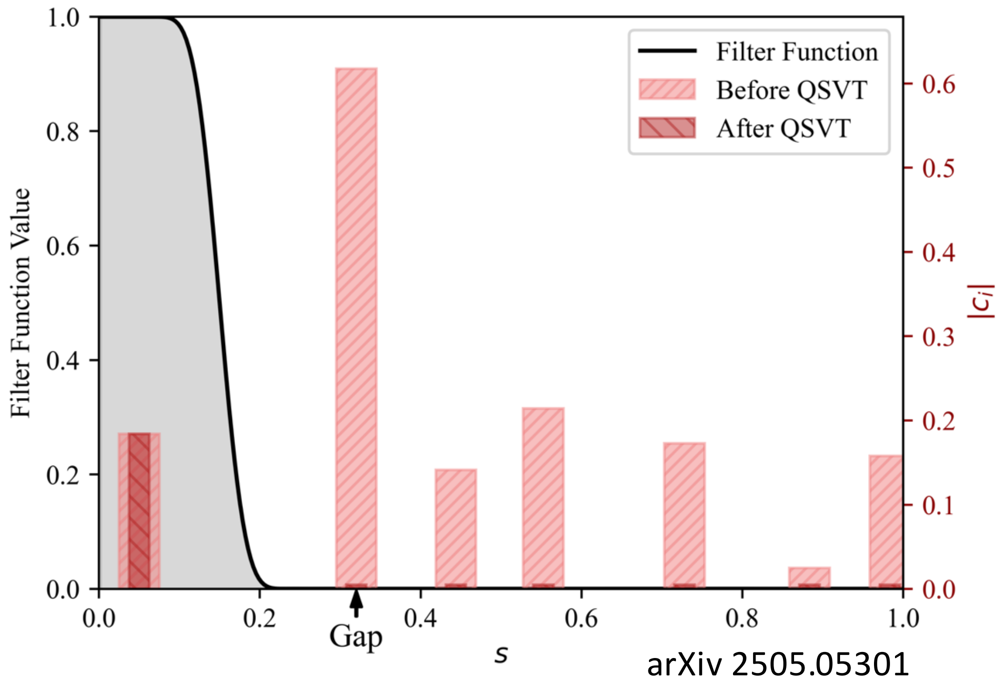

arXiv 2505.05301

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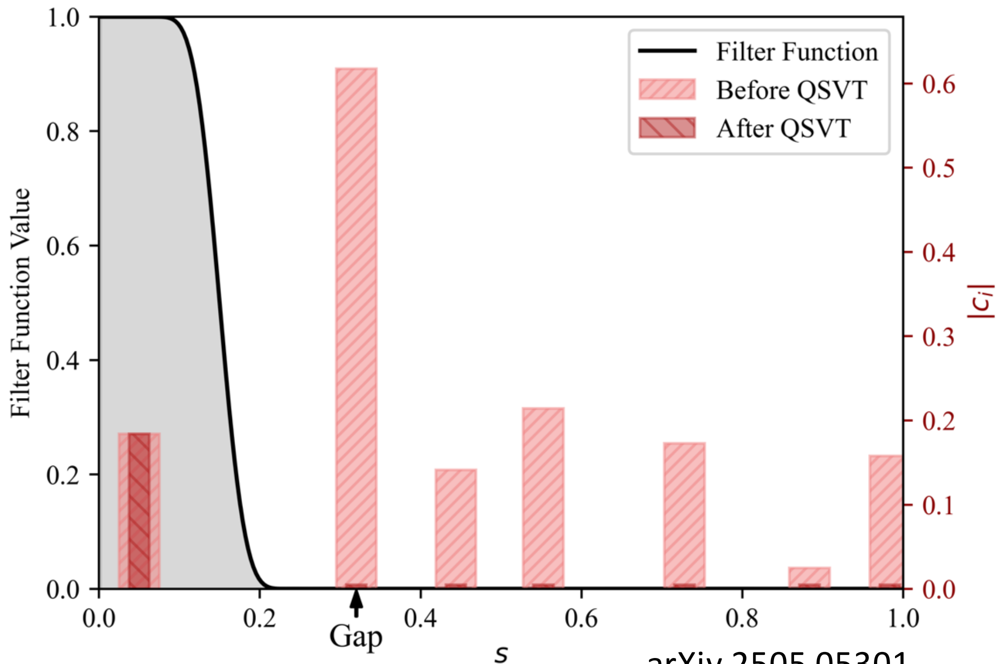

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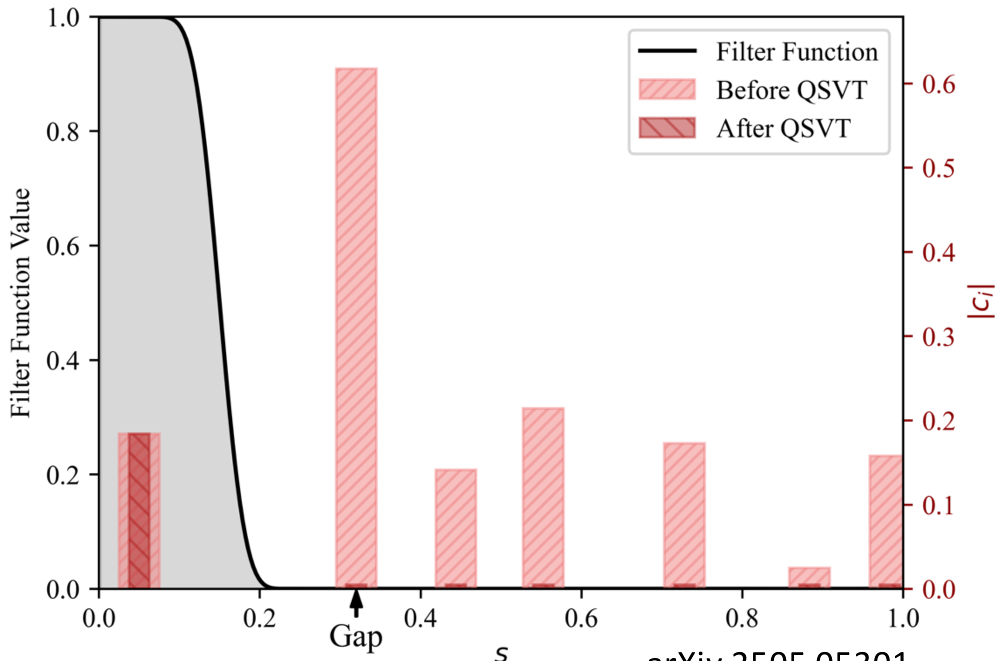

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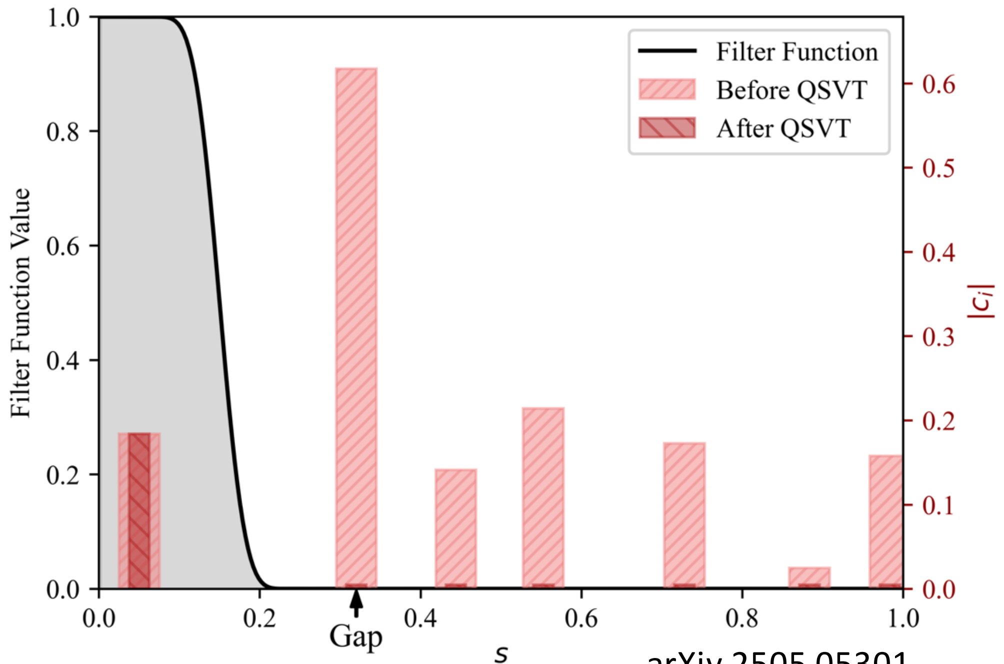

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Discrete versus continuous operator

- Witten Laplacian: $\mathcal{H} \;:=\; -e^{\beta E/2}\mathcal{L}e^{-\beta E/2}$

$$\mathcal{H} = \sum_{j=1}^{d} L_j^\dagger L_j, \quad L_j := -i\frac{1}{\sqrt{\beta}}\partial_{x_j} - i\frac{\sqrt{\beta}}{2}\partial_{x_j}E \quad \forall j \in [d] = \{1,2,\ldots,d\}$$

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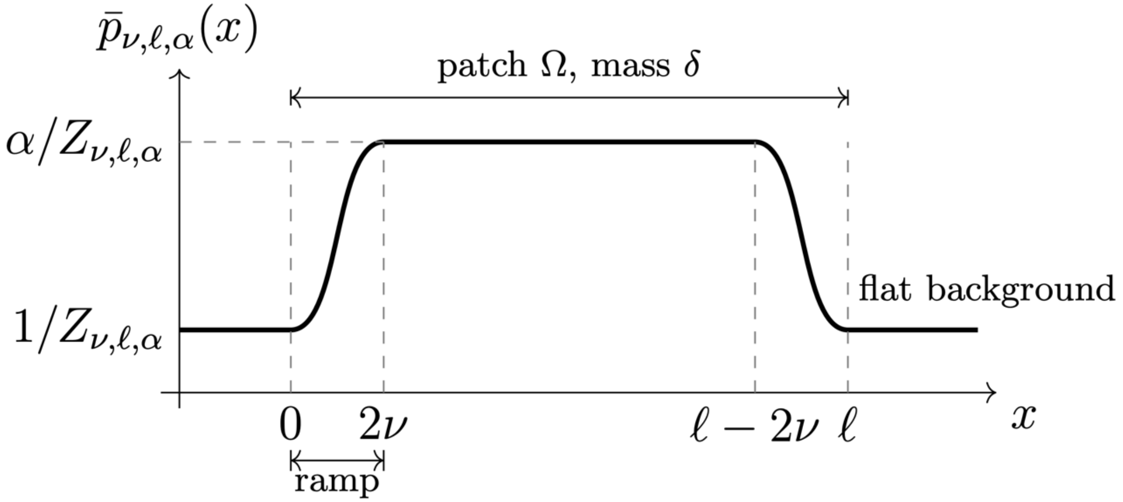
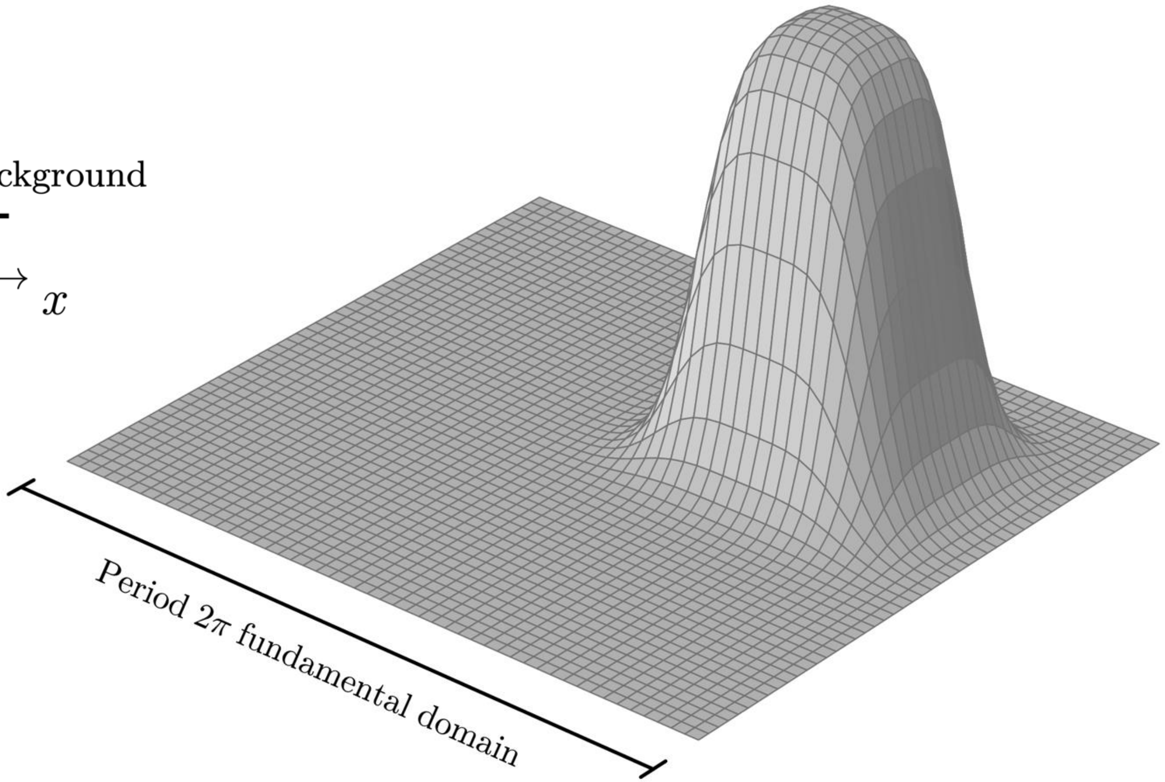
Classical Gibbs sampling in $\Omega(e^{\beta\Delta})$

- OSHHKSR26: Gevrey potentials can hide $O(e^{\beta\Delta})$ bump functions from classical algorithms.





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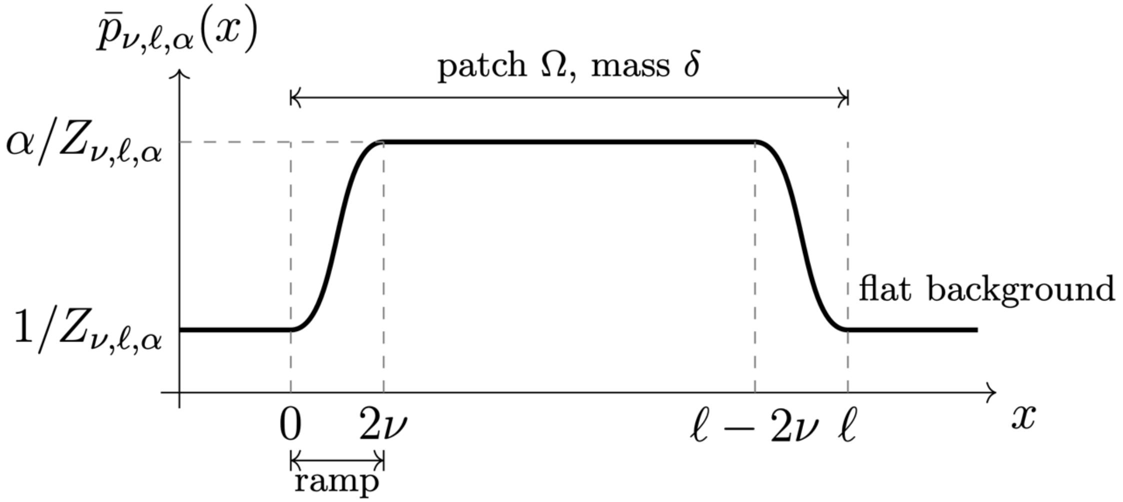
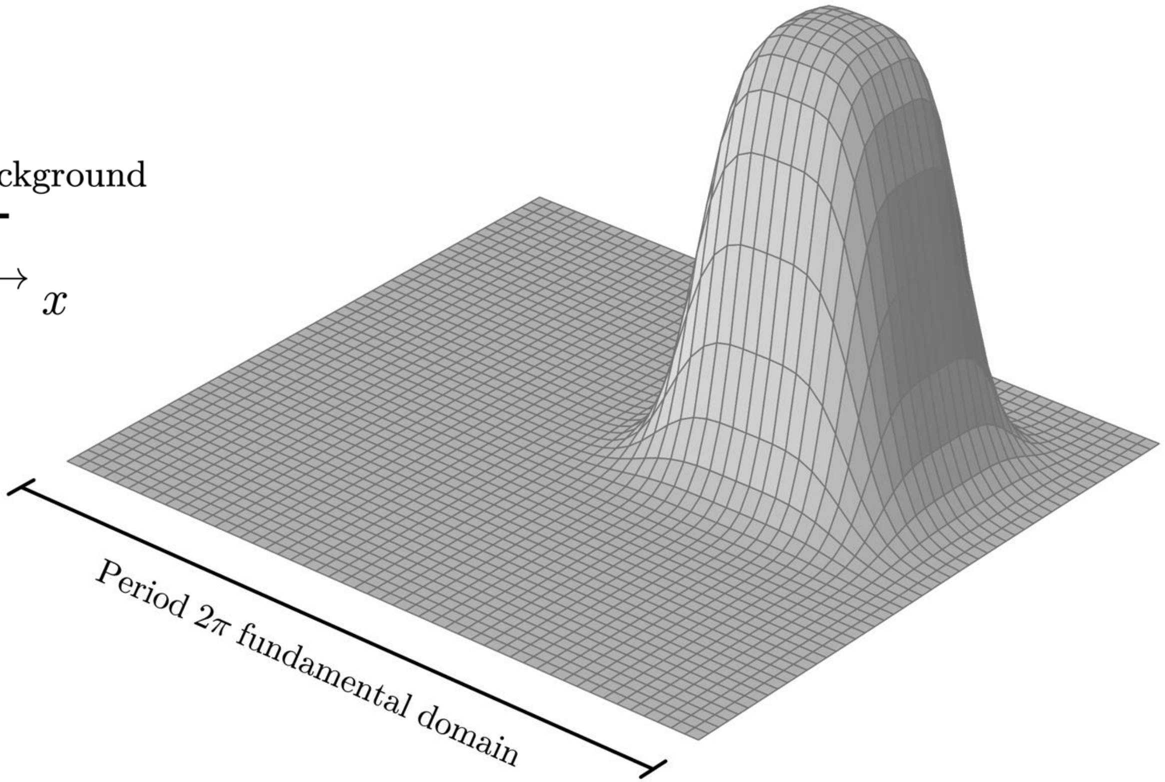




- Comparing the quantum upper-bound and the classical lower bound provides a provable quadratic quantum—classical separation.

Application 3: Force-field molecular dynamics

Force field as a function of torsion angles:

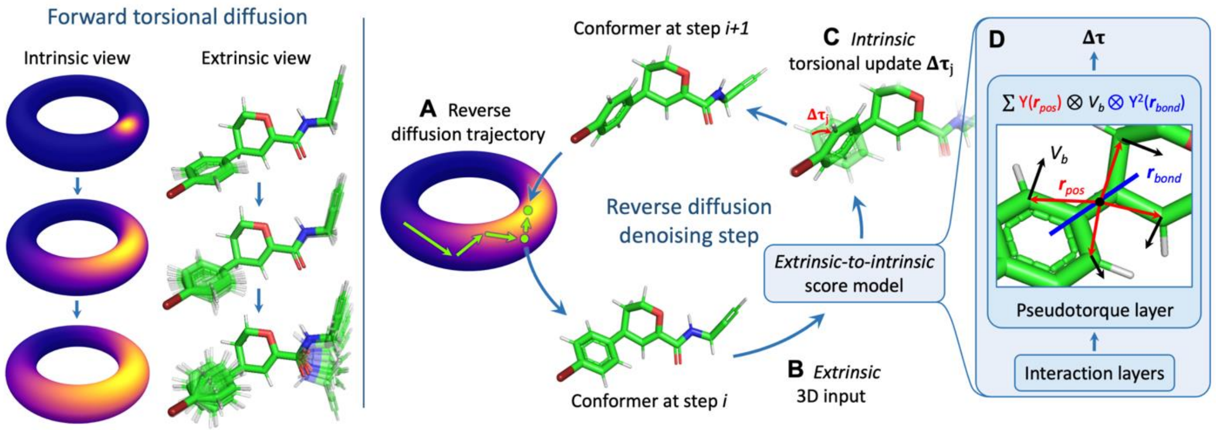
$$U = \sum_{\text{bonds}} \frac{1}{2} k_b (r - r_0)^2 + \sum_{\text{angles}} \frac{1}{2} k_a (\theta - \theta_0)^2 + \sum_{\text{torsions}} \frac{V_n}{2} [1 + \cos(n\phi - \delta)]$$

$$+ \sum_{\text{improper}} V_{imp} + \sum_{\text{LJ}} 4\epsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^{6}}{r_{ij}^{6}} \right) + \sum_{\text{elec}} \frac{q_i q_j}{r_{ij}},$$

36th Conference on Neural Information Processing Systems (NeurIPS 2022).

Torsional Diffusion for Molecular Conformer Generation

Bowen Jing,*[1] Gabriele Corso,*[1] Jeffrey Chang,[2] Regina Barzilay,[1] Tommi Jaakkola[1]
[1]CSAIL, Massachusetts Institute of Technology [2]Dept. of Physics, Harvard University



Conclusions

- Gevrey smoothness is a natural criteria for efficient quantum computing on continuous functions.
- QLSA solves general linear PDEs with periodic boundary conditions with no curse of dimensionality for the Gevrey class.
- QSVTh enables high-precision Gibbs sampling for Gevrey potentials.
- Gevrey potentials provide quadratically more adversarial examples for any classical algorithms thereby proving a quantum-–classical separation.
- Many more applications to yet to be explored: Molecular dynamics, time-dependent PDEs, etc.

Thank you for your attention!

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